

Remote management of critical care patients

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Management Team



Chayan Chatterjee

- Director, MedTech – Glocal Healthcare
- Product Management, Sales Operations – OmniGuide Surgical (MA, USA)
- Factory Operations – ITC Limited
- B.Tech, M.Tech (IIT Bombay), MBA (Wharton)



Sahil Mehta

- Sr. Product Manager, MedTech – Glocal Healthcare
- Device Development – USA, India
- Co-founder, Delhi Makers
- BS (Univ. of Utah), M.Eng (Cornell)



Cornell University



Soura Bhattacharyya

- CTO, HR Head – Glocal Healthcare
- Director, Business Development – OmniGuide Surgical (MA, USA)
- Project Management – ITC Limited
- B.Tech (IIT KGP), MBA (Harvard)



HARVARD BUSINESS SCHOOL

About Lattice



- Incorporated in November 2014, Debt financing from friends and family
- Current team size of 16, across software development, embedded systems engineering and product sales

Product Development

- 4 distinct offerings commercialized till date
 - Networked ICU patient monitor
 - SMS-enabled patient monitor
 - Cardiac risk assessment kiosk
 - Bluetooth-enabled spirometer

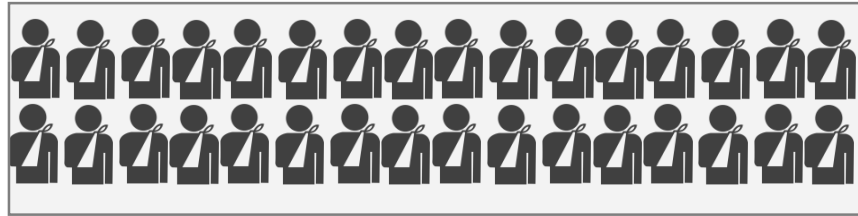
Design Services

- Project based engagement
 - Need discovery
 - Design & Prototyping
 - Build for manufacture
- 8 projects completed, for range of customers from healthcare providers, medtech startups, medical device manufacturers to healthcare technology consortiums

- Turnover **exceeding 2.4Cr** over a period of two years **(2015-17)** – 85/15 – Design Services/Product Sales

Strategy: **Design Services** as a conduit for need identification, **Products** to achieve scale

Critical Care Challenges in India



Surge in volume and severity of cases treated

Non-communicable diseases, infectious diseases, trauma, poisonings and envenomations



Inadequate ICU infrastructure

70,000 beds across all hospitals and nursing homes cater to 5 million patients requiring ICU care each year

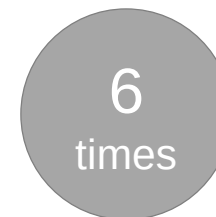


Deficiency in availability of skilled human resources

- Number of trained intensivists/ acute care specialists in the low hundreds
- Estimated 35% shortage of intensivists by 2020 as a result of the aging population and the growing demand for intensive care services



is what 46% of patients travel to seek medical care

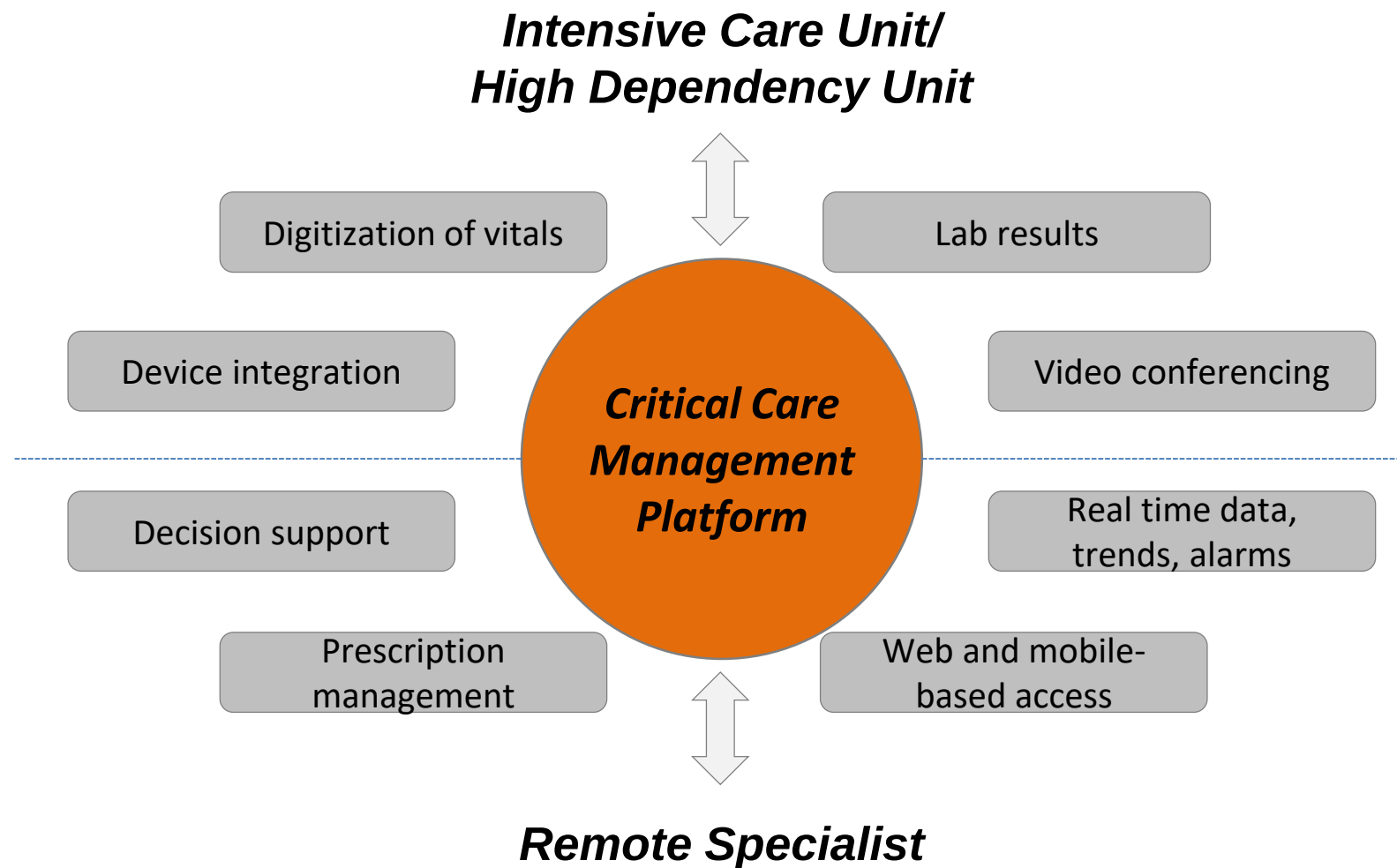


more doctors per capita in urban India, compared to rural

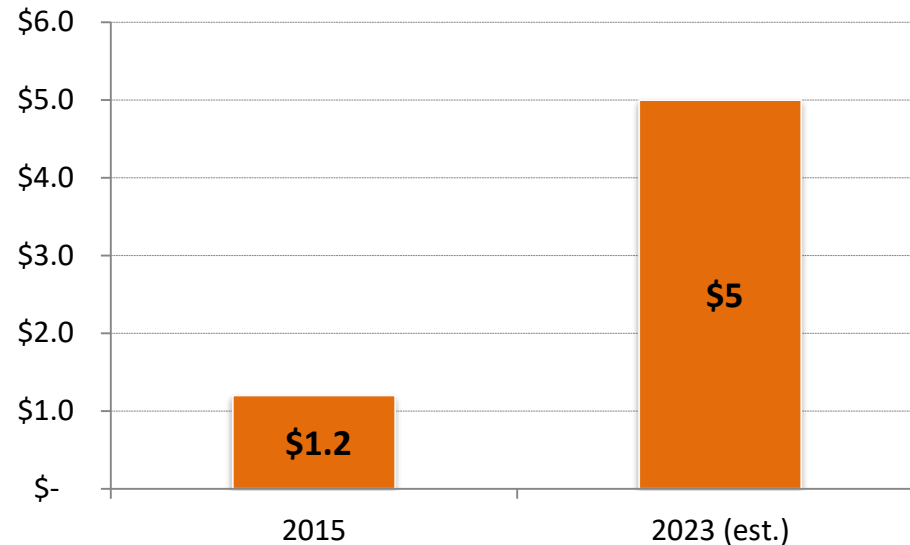


of the population driven to poverty due to healthcare expenditure

The Solution: Remote Critical Care Management



Market Assessment



Global Tele-ICU market size (\$ billion)

Key findings

- Asia-Pacific tele-ICU market (~ 20%) size expected to grow faster than 23%
- Hardware components comprised of over 80% share
- Software components predicted to grow rapidly during the forecast period

Growth drivers

- Clinical evidence indicating reduced clinical decision time, complication rates and length of stay
- Improvements in telecom infrastructure

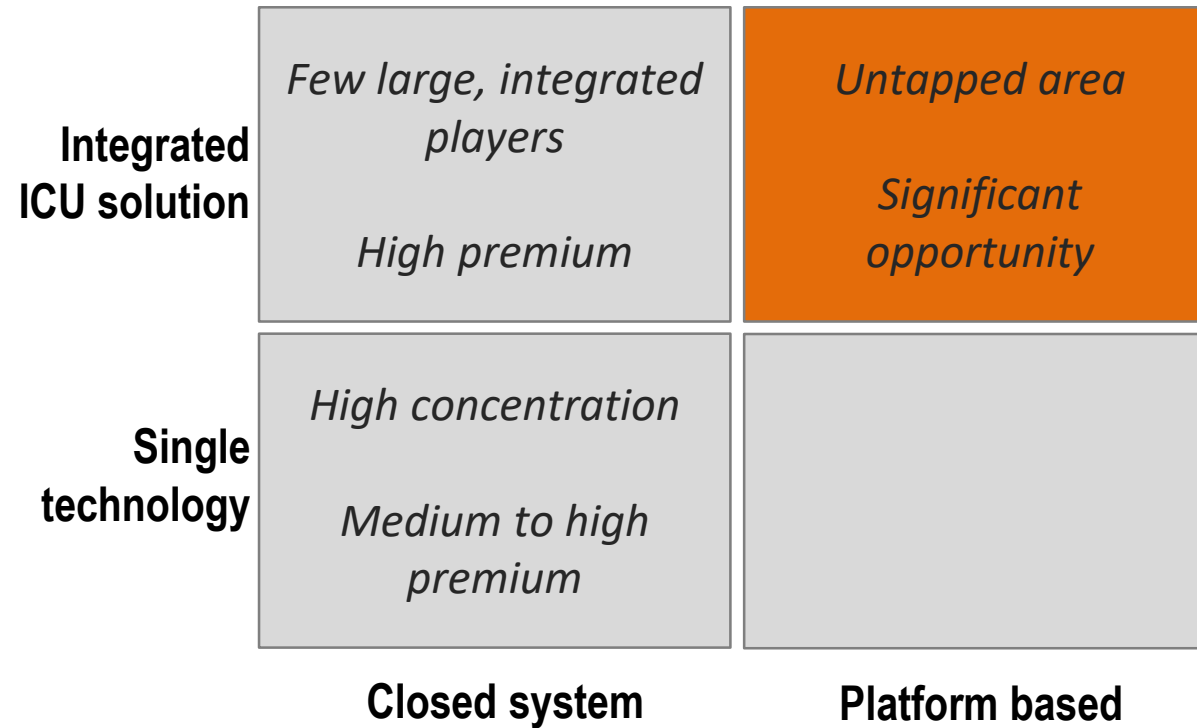
Constraints

- High setup costs
- Lack of organized service providers

Activity in India

Fortis Critinext, InTeleICU, Medica

Competition

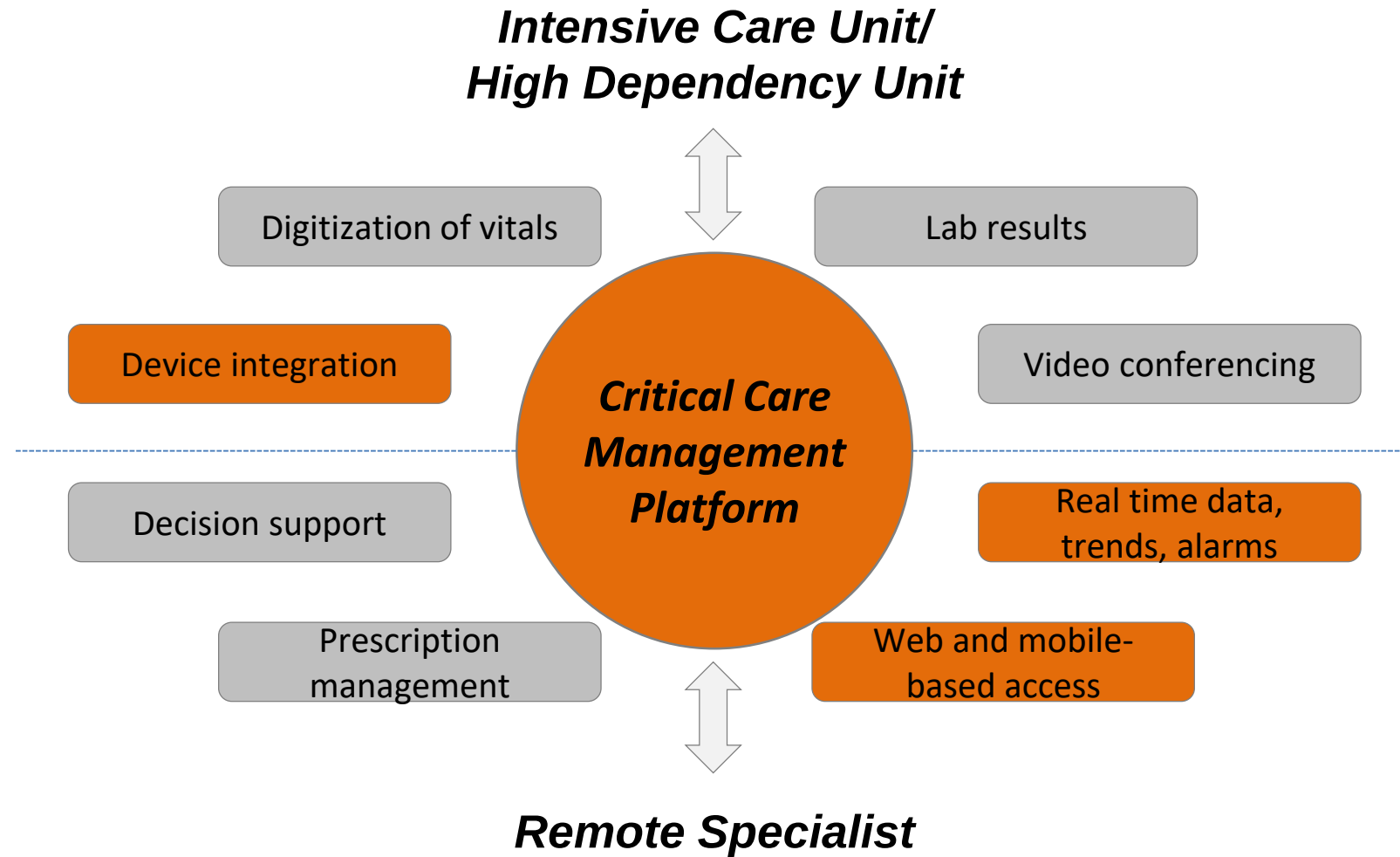


Current teleICU systems limited by –

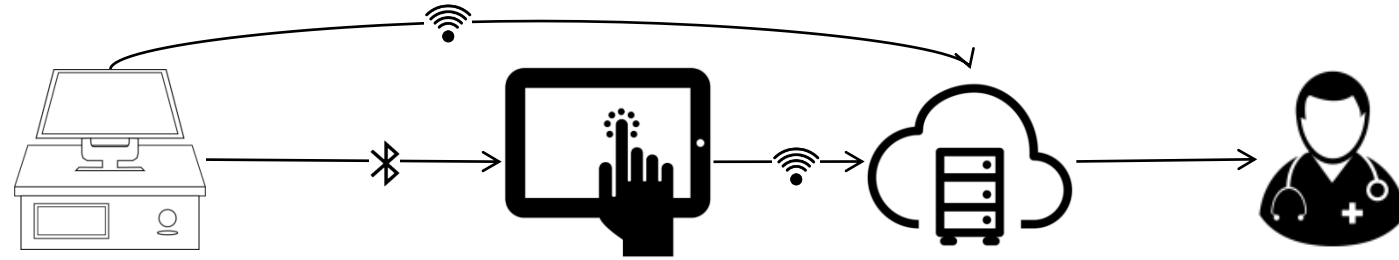
1. Closed design offering little or no flexibility to adopt alternate brands or utilize existing equipment
2. Few players providing entire solution for an ICU use case, characterized by extremely high setup and operating costs



Development Status



Device Integration: Current Design



- Proprietary design – Lattice FYVE S1
- Plug and play device with built in wireless and bluetooth modules
- Multipara Monitor – ECG, SpO2, Temperature, NIBP, Respiration Rate
- Data transmission using 2G/3G/Broadband internet

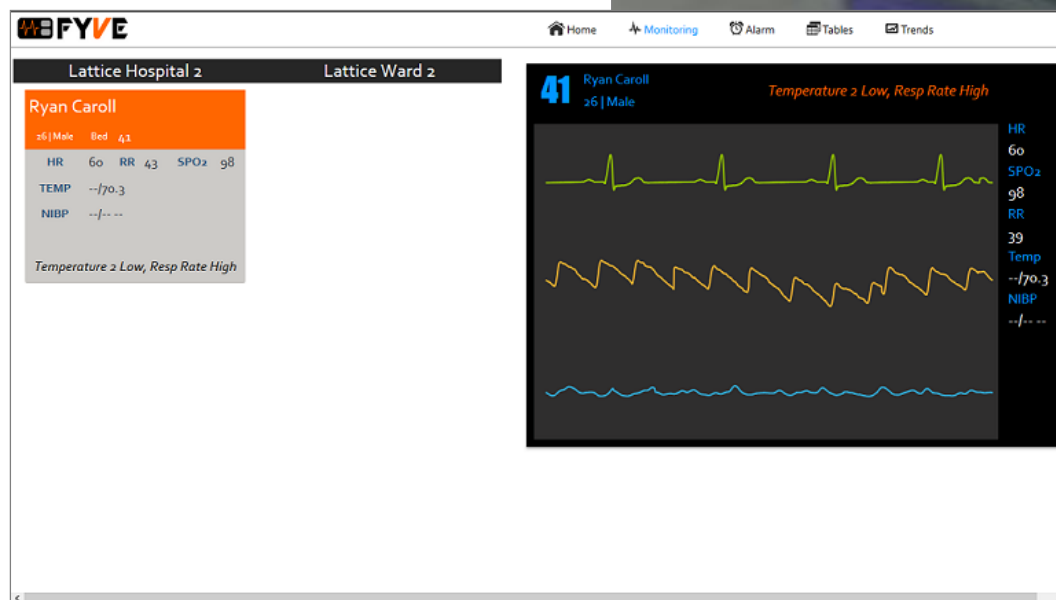
- Touch screen display. Provision for multilingual User Interfaces
- Standardized as well as free text data entry
- Easy integration with EMR
- Video conferencing using built in camera
- Rapid product line extensions – data transfer using GSM, GPRS modules for a wide range of settings

- Mobile/ Web-based real time access to patient details
- Vital statistics, waveforms, alarm history and trend data
- Facilitates Decentralized Monitoring – lower capital expenditure, improved utilization of clinical resources

Device Integration: Current Design



FYVE S1 patient monitor
with tablet-based display



Web-based Remote
Monitoring Console



Sales

- Networked patient monitor for ICU use – Makunda Christian Hospital (2 units), Baylor College of Medicine (1 unit)
- SMS-enabled patient monitoring system for PHC use – 40 units sold for project commissioned by National Health Systems Resource Center (NHSRC)

Pilots

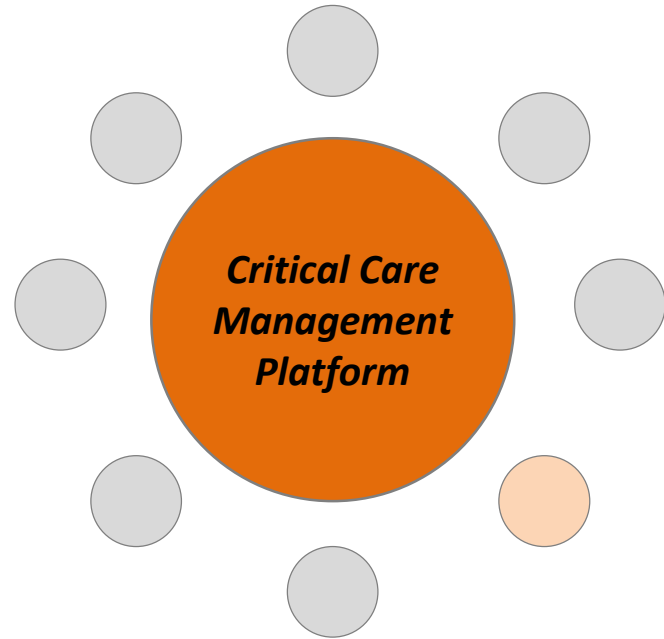
- Successful trials in CMC Ludhiana, NKS Hospital (Delhi & Panipat), Sanjay Gandhi Hospital (Amethi), Baptist Christian Hospital (Assam), Garg Hospital (Noida) and Prakash Hospital (Noida)
- Ongoing pilot at Max Hospital (Saket) and upcoming pilot at Apollo Hospital (Hyderabad)

Awards and Recognition

- Recipient of the 6th BIRAC biotechnology ignition grant (BIG) award in 2015, for innovative work in developing affordable tele-ICU solutions.
- Top Three startup at Nasscom Product Conclave
- Top Five startup at BioAsia 2017
- Hot 100: Race to Grace Award 2017



Design Integration: Future Roadmap



- OEM devices for ICU usage – Patient monitors, Ventilators, Syringe pumps, Infusion monitors etc.
- Lattice FYVE S1 Multipara Patient Monitor

- Integration of other devices used in ICUs. Curated list of best-in-class technologies
- Platform based approach – Encourage participation by medical device OEMs
- One-stop-shop for an ICU setup, with flexibility in choice of device manufacturer/ technology
- Overcomes major shortcoming of closed systems offered by current teleICU players



Platform-based approach: Plug-and-play system - Industry-standard software/ hardware interfaces that can accommodate multiple devices



Integrated decision support tools – Standardized ICU Guidelines

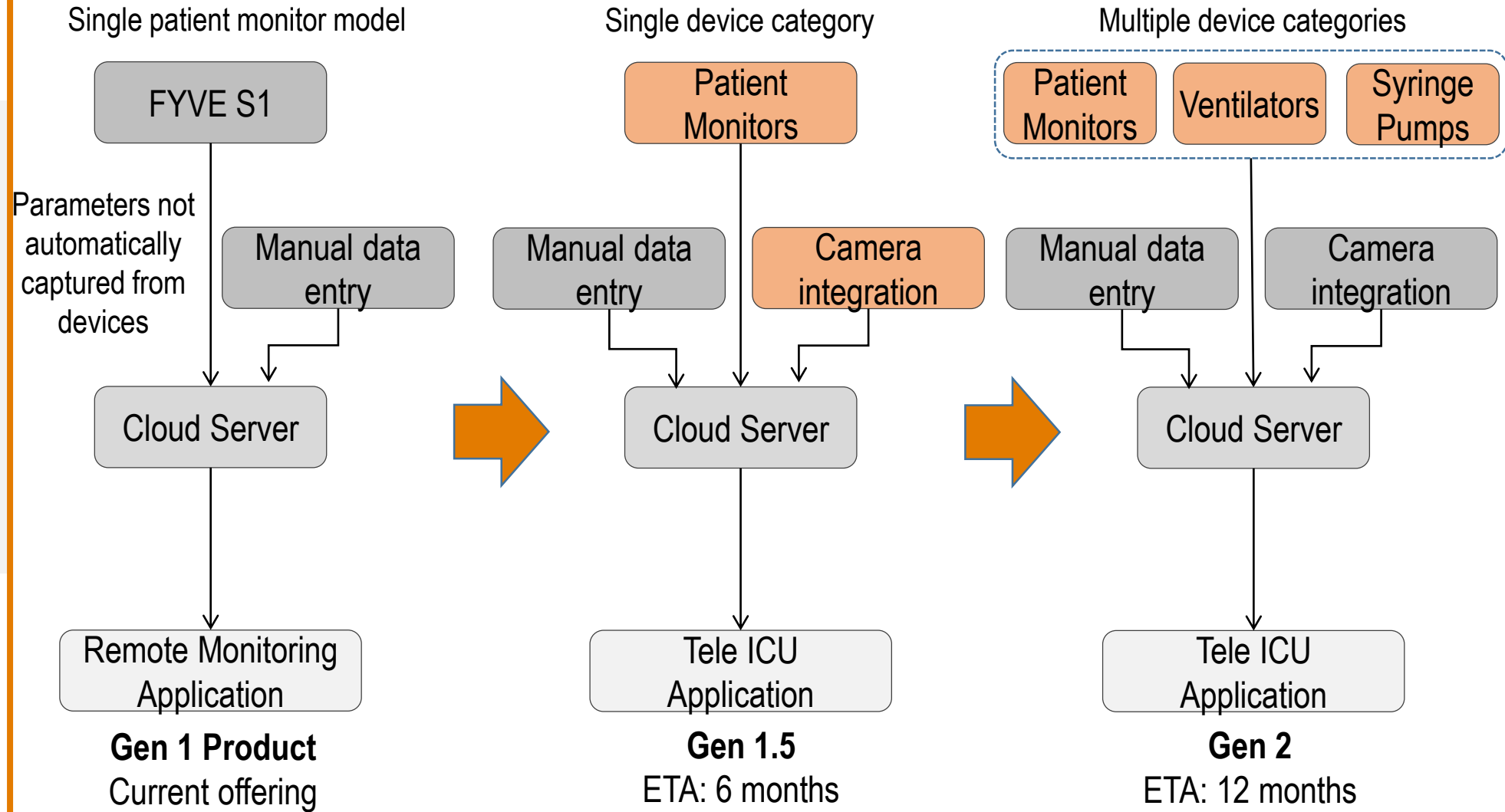


Wireless connectivity makes the system plug-and-play, allowing it be installed in various settings – hospitals, at home, in ambulance. Custom adapter to be developed for non-wireless devices



Designed for data portability: HL7 FHIR implementation for easy interfacing with HMIS systems

Product development roadmap



Business Model - Domestic & International



Customer Segment	Engagement Model
- Hospitals, Nursing homes - Government run - For profit	- Lattice sales team - Clinical partners, e.g. Fortis Critinext - Distribution partners (international)
At home care	Home healthcare providers, e.g. HCAH
In ambulance patient monitoring	- Lattice sales team - Ambulance aggregators, e.g. Ambee - Distribution partners (international)

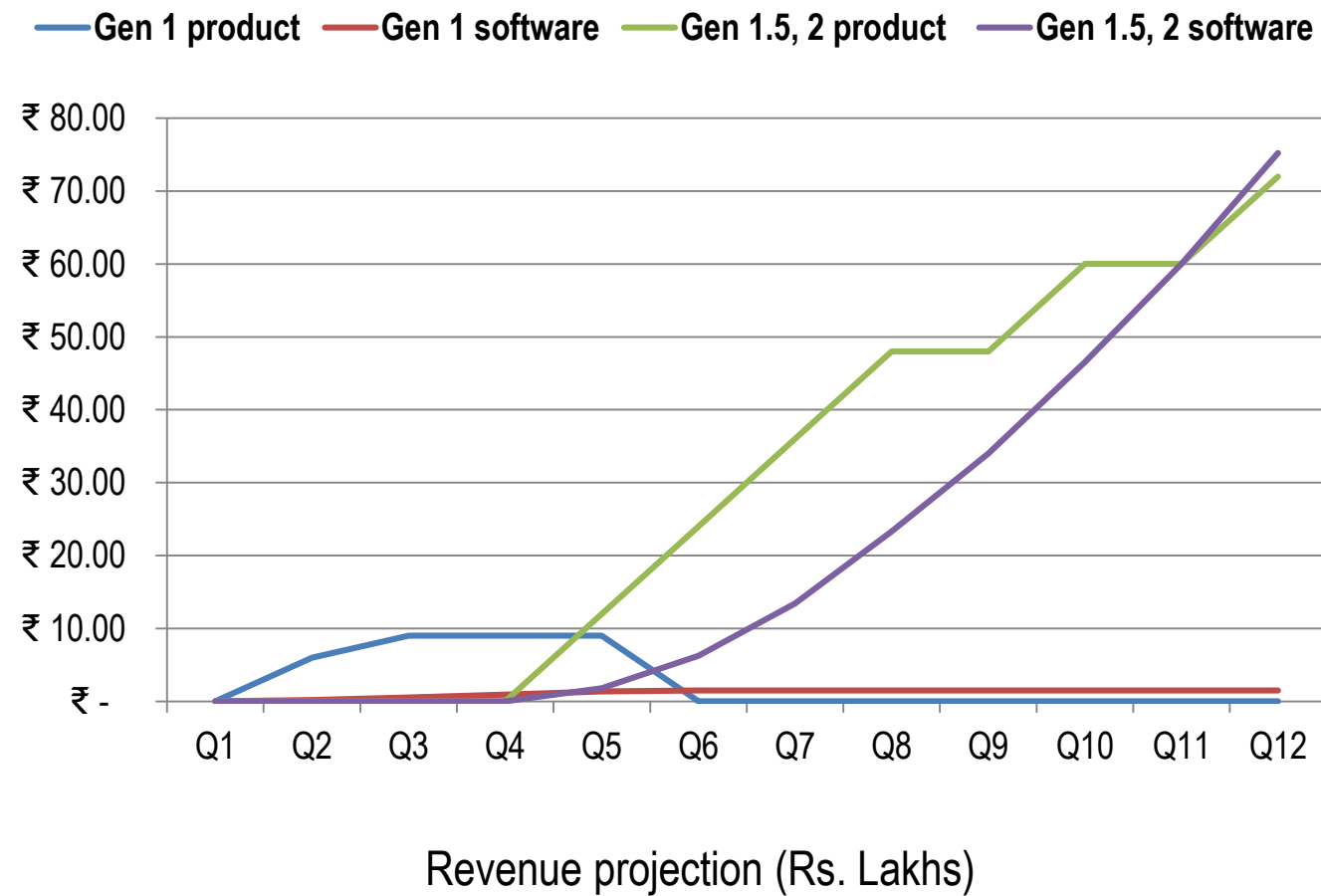
Pricing strategy

- Hardware
 - Direct, one time CAPEX per sale
 - Annual maintenance charges for device maintenance
- Software – Pay per use (per patient per day of use)

Unit Economics

- Hardware
 - Own brand: 50-60% gross margin
 - OEM brand: 15-20% gross margin
- Software
 - Critical Care Management Platform: Rs. 99 per patient per day (pay-per-use)
 - 10 patient days per month (per device) to achieve breakeven

Financial Projections





Achievements



Certification

- Registration under Medical Device Quality Management System (ISO 13485:2003) for Manufacture and Supply of Diagnostic Devices, Surgical Tools and Monitoring Devices

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Pilots

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